

Problem

Sketch Bode magnitude and phase plots for

$$H(s) = \frac{1.6}{s(s^2 + s + 16)}, \quad s = j\omega$$

Step-by-step solution

Step 1 of 10

Consider the following transfer function:

$$G(s) = \frac{1.6}{s(s^2 + s + 16)} \dots\dots (1)$$

Substitute $j\omega$ for s .

$$\begin{aligned} G(j\omega) &= \frac{1.6}{j\omega[(j\omega)^2 + j\omega + 16]} \\ &= \frac{1.6}{16j\omega\left[1 + \frac{j\omega}{16} + \left(\frac{j\omega}{4}\right)^2\right]} \\ &= \frac{0.1}{j\omega[1 + 0.0625j\omega - 0.0625\omega^2]} \end{aligned}$$

The system has three poles, one at origin, and two complex poles at $-0.5 \pm j3.97$.

Step 2 of 10

Determine the magnitude of the constant.

$$20 \log(0.1) = -20 \text{ dB}$$

The pole at origin has a magnitude of -20 dB/decade and the complex poles at $\omega_n = 4 \text{ rad/s}$ have a slope of -40 dB/decade .

Magnitude plot:

Compare equation (1) with standard form.

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = s^2 + s + 16$$

$$\omega_n^2 = 16$$

$$\omega_n = 4$$

Choose a low frequency ω_l such that $\omega_l < \omega_n$ and choose a high frequency ω_h such that $\omega_h > \omega_n$.

Step 3 of 10

Consider $\omega_l = 0.1 \text{ rad/s}$ and $\omega_h = 100 \text{ rad/s}$

Calculate the magnitude at lower cut-off frequency $\omega = \omega_l$.

Write the expression for magnitude at lower cut-off frequency $\omega = \omega_l$.

$$\begin{aligned} A &= 20 \log \left| \frac{0.1}{j\omega} \right| \\ &= 20 \log \left| \frac{0.1}{0.1} \right| \\ &= 0 \text{ dB} \end{aligned}$$

Therefore, the value of the magnitude at lower cut-off frequency $\omega = \omega_l$ is 0 dB.

Step 4 of 10

Calculate the magnitude at natural frequency $\omega = \omega_n$.

Write the expression for magnitude at natural frequency $\omega = \omega_n$.

$$\begin{aligned} A &= \left\{ 20 \log(0.1) + 20 \log \left| \frac{1}{j\omega} \right| \right\} \bigg|_{\omega=4} \\ &= 20 \log(0.1) + 20 \log \left| \frac{1}{(4)} \right| + \\ &= -20 + 20 \log \left| \frac{1}{(4)} \right| \\ &= -20 + (-12.04) \\ A &= -32.04 \text{ dB} \end{aligned}$$

Therefore, the value of the magnitude at natural frequency $\omega = \omega_n$ is -32.04 dB.

Step 5 of 10

Calculate the magnitude at upper cut-off frequency $\omega = \omega_h$.

Write the expression for magnitude at upper cut-off frequency $\omega = \omega_h$.

$$\begin{aligned} A &= 20 \log \left| \frac{0.1}{j\omega} \right| \\ &= 20 \log \left| \frac{0.1}{100} \right| \\ &= -60 \text{ dB} \end{aligned}$$

Therefore, the value of the magnitude at upper cut-off frequency $\omega = \omega_h$ is -60 dB.

Step 6 of 10

Sketch the slopes of the individual terms of the function as shown in Figure 1.

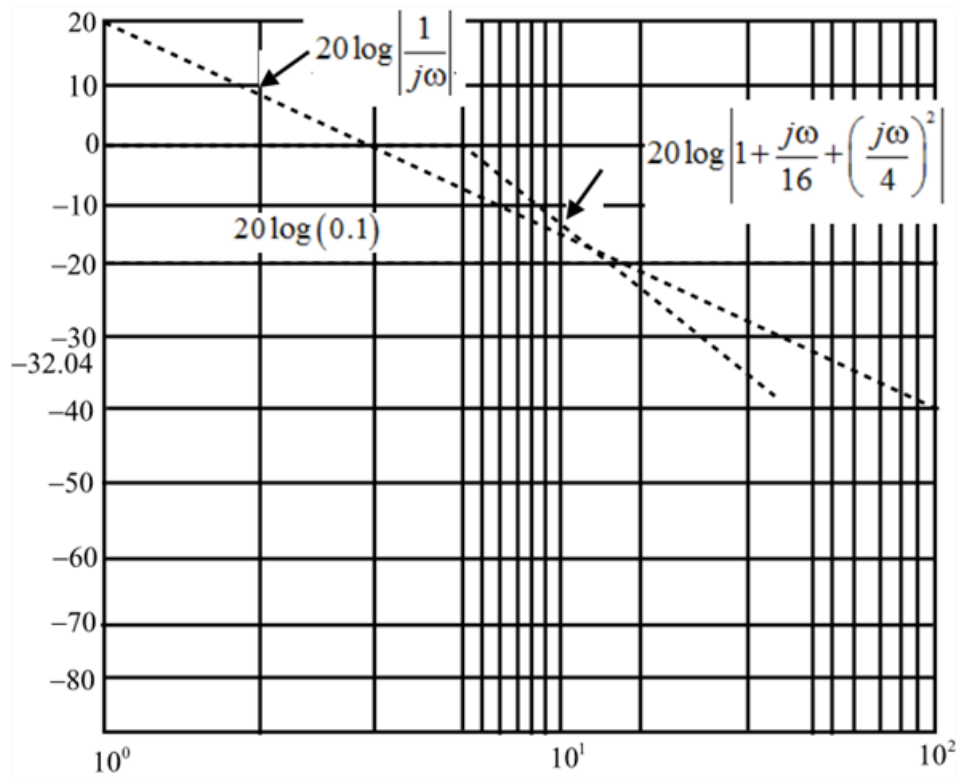


Figure 1

Step 7 of 10

From Figure 1, the overall magnitude plot is as shown in Figure 2.

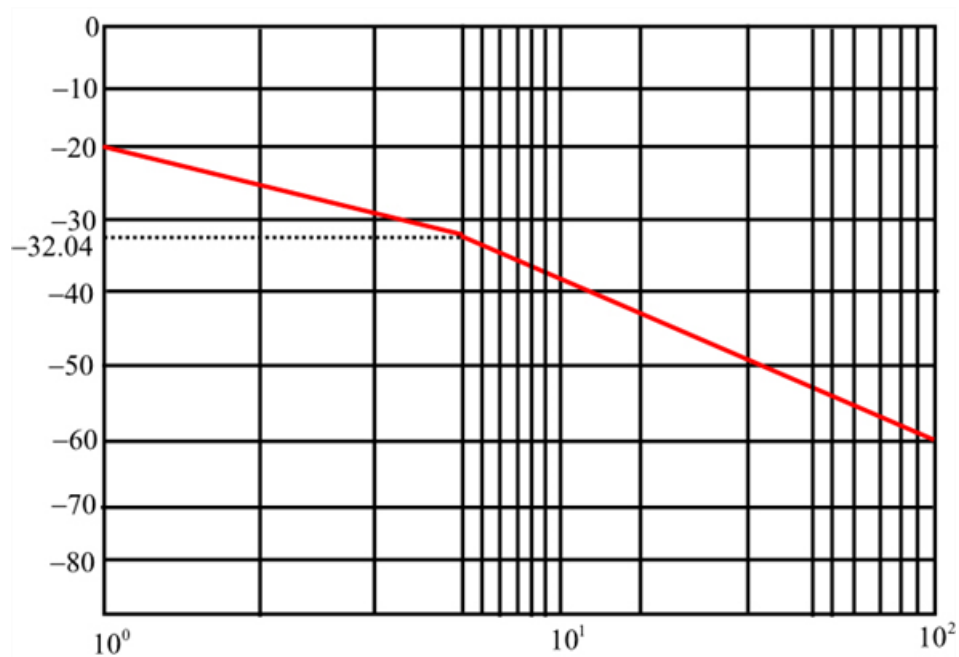


Figure 2

Step 8 of 10

Phase plot:

Write the phase angle of $G(j\omega)$ as a function of ω .

$$\begin{aligned}\phi &= \angle G(j\omega) \\ &= -90^\circ - \left[\tan^{-1} \left(\frac{0.0625\omega}{1 - 0.0625\omega^2} \right) \right]\end{aligned}$$

For $\omega > \omega_n$,

$$\begin{aligned}\phi &= \angle G(j\omega) \\ &= -90^\circ - \left[\tan^{-1} \left(\frac{0.0625\omega}{1 - 0.0625\omega^2} \right) + 180^\circ \right] \text{ For } \omega > \omega_n\end{aligned}$$

The phase angle of $G(j\omega)$ are calculated for various values of ω .

Table 1

ω rad/s	$\phi = \angle G(j\omega)$
0.1	-90.35°
4	-180°
50	-268.85°

The phase of the constant is zero.

The phase of pole at origin is -90° .

The phase of the complex poles is -180° .

Step 9 of 10

The plot of these terms is as shown in Figure 3.

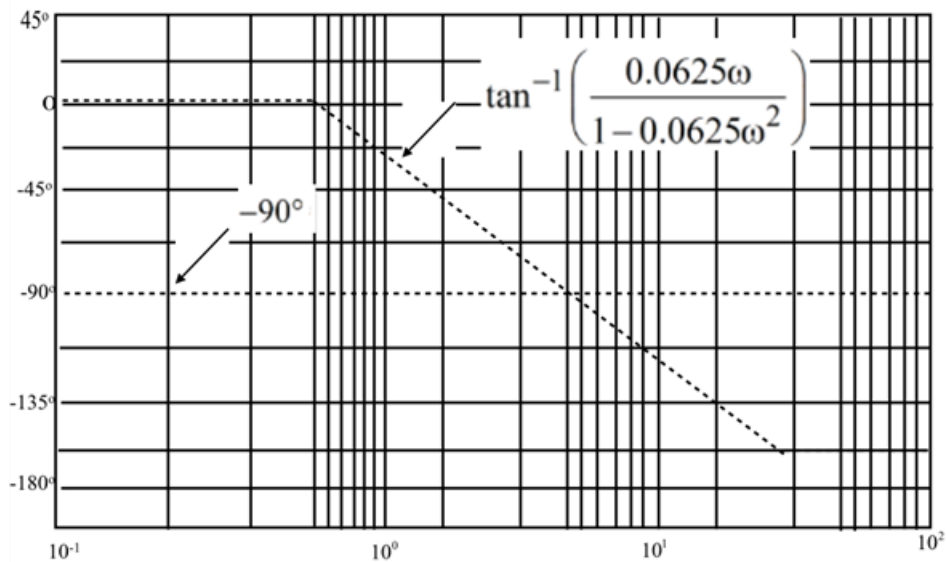


Figure 3

Step 10 of 10

Sketch the asymptotic phase plot for the system.

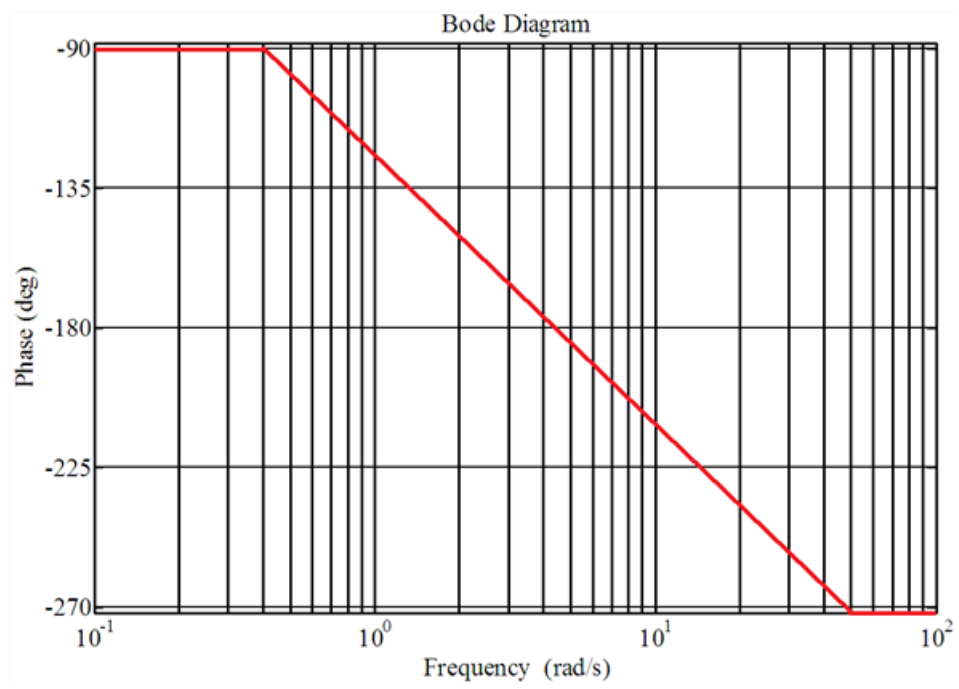


Figure 4

Thus, the Bode magnitude and phase plots for the system are sketched.